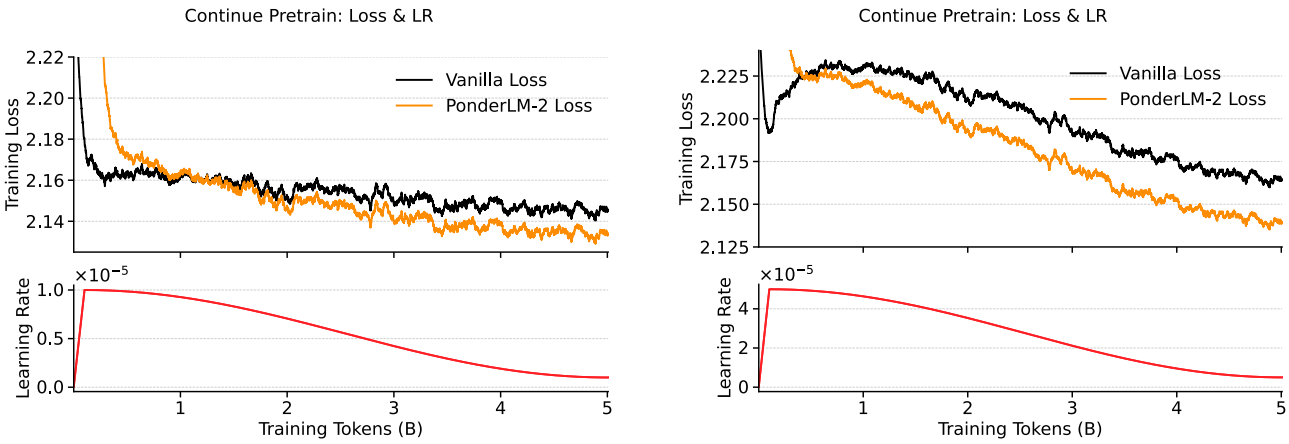

(a) Peak LR = 1e-5. *(b)* Peak LR = 5e-5.

Figure 4. **Learning rate ablation for continual pretraining on LLaMA-3-3B.** Training loss of Vanilla CPT and PonderLM-2 CPT on SlimPajama (5B tokens) under two peak learning rates. PonderLM-2 exhibits a small transient loss spike at the start as it accommodates the newly introduced latent states, then quickly recovers and surpasses Vanilla CPT. Both learning rates yield stable training with no recurring spikes. Higher LR (5e-5) leads to faster crossover but slightly higher final loss; lower LR (1e-5) provides smoother convergence and lower final loss.

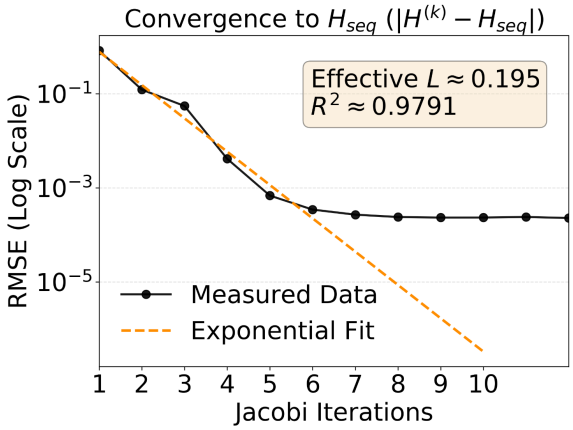

Figure 5. **Jacobi iteration convergence of the LLaMA-3-3B CPT model.** RMSE between the hidden states at iteration k and the sequential fixed point H_{seq}, plotted on a log scale. The effective Lipschitz constant is $L \approx 0.195$ ($R^2 \approx 0.979$), indicating rapid exponential convergence: RMSE drops by over two orders of magnitude within 6 iterations and plateaus thereafter.